FundMe v1.0 Security Audit Report

Version 1.0

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Protocol Summary

FundMe v1.0 is a fully decentralized, autonomous crowdfunding protocol that leverages smart contracts for transparency, Chainlink Oracles for accurate USD-to-ETH price conversions, and Chainlink Automation to handle campaign finalization without human intervention. Donors receive unique NFT Tier Rewards stored on IPFS as immutable proof of support.

The protocol operates through a state-machine with the following lifecycle:

1. **NOT_OPEN_YET** - Initial state after deployment
2. **OPEN** - Active fundraising phase
3. **CLOSED** - Campaign ended (target reached or deadline passed)
4. **FINALIZED** - Owner has withdrawn funds

Key features include:

- Minimum $5 USD contribution requirement (via Chainlink ETH/USD price feed)
- Tiered NFT rewards: Bronze ($50+), Silver ($200+), Gold ($1000+)
- Chainlink Automation for automatic campaign closure

Disclaimer

The raihanmd team makes all effort to find as many vulnerabilities in the code in the given time period, but holds no responsibilities for the findings provided in this document. A security audit by the team is not an endorsement of the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the Solidity implementation of the contracts.

Risk Classification

		Impact		
		High	Medium	Low
Likelihood	High	CR	H	M
	Medium	H	M	M/L
	Low	M	M/L	L

We use the CodeHawks severity matrix to determine severity. See the documentation for more details.

Audit Details

- Commit Hash: cc06172ede346cf94f44780a267e7a6c95a7e64c
- Solc Version: ^0.8.19
- Chain(s) to deploy contract to: Ethereum (Sepolia tested)

Scope

- In Scope:

```
./src/
  FundMeRewardNft.sol
  NftTier.sol
  PriceConverter.sol
  CampaignLifeCycle.sol
  CampaignClosedReason.sol
```

Roles

There are 3 types of actors in the protocol:

Actors:

1. Owner

- deploy the smart contract
- `openCampaign` to start fundraising
- `closeCampaign` (also callable by anyone, as per README)
- `withdraw` funds after campaign closes
- `claimRewardNft` on behalf of funders (via FundMeRewardNft)
- modify NFT tier URIs

2. Funders

- `fund` ETH to active campaigns
- `claimRewardNft` based on their contribution tier

3. Chainlink Automation / Anyone

- can call `performUpkeep` to trigger automatic campaign closure
- can call `closeCampaign` directly

Executive Summary

This security audit of FundMe v1.0 identified **11 issues** across multiple severity levels. The most critical finding is the NFT tier eligibility calculation occurring at claim time rather than fund time, which leads to users receiving NFTs they weren't originally entitled to based on ETH price volatility. Another significant issue is the owner's ability to modify NFT metadata after minting, creating a potential rug vector.

Issues found

Severity	Number of issues found
CRITICAL	1
HIGH	0
MEDIUM	1

Severity	Number of issues found
LOW	6
INFO	3
TOTAL	11

Test Coverage

All findings have corresponding Foundry tests in `test/audit/Audit.t.sol`:

Finding	Test Function	Status
CR-1	`test_CR1_TierDilution_PriceIncreaseAfterFunding`	PASS
M-1	`test_M1_CampaignCanOpenWithOneSecondDuration`	PASS
M-1	`test_M1_GriefingWithInstantCampaign`	PASS
L-1	`test_L1_DocumentAddressZeroLimitation`	PASS
L-2	`test_L2_DocumentGasIssue`	PASS
L-3	`test_L3_OwnerCanWithdrawAllFunds`	PASS
L-4	`test_L4_DocumentGasWaste`	PASS
L-6	`test_L6_OwnerCanChangeTierUriAfterMint`	PASS
L-6	`test_L6_FundMeOwnerCannotChangeUri`	PASS
I-3	`test_I3_FrontRunFundWithCloseCampaign`	PASS
I-3	`test_I3_MultipleUsersCanRaceToClose`	PASS

Findings

Critical

[CR-1] NFT Tier Eligibility Calculated at Claim Time, Not Fund Time

Description:

The NFT tier determination in `FundMe.claimRewardNft` uses the current ETH/USD price at claim time rather than the price at funding time. If ETH price increases significantly between funding and claiming, a user who funded $49 USD worth of ETH could become eligible for a Bronze NFT even though their original contribution was below the $50 threshold.

```
uint256 totalFundedEth = s_roundToAddressToAmountFunder[roundCampaign][
    msg.sender];
uint256 totalFundedUsd = totalFundedEth.getConversionRate(s_priceFeed);
    // @>

if (totalFundedUsd < BRONZE_THRESHOLD) { // @>
    revert FundMe__BalanceHaveToAtLeastEqualBronzeThreshold();
}
```

Impact: Critical

- Users may receive NFTs they weren't entitled to based on their original contribution, diluting the NFT value for legitimate tier holders
- Early funders who claimed during low ETH prices may have received higher-tier NFTs while later funders in the same tier receive lower-tier NFTs
- The economic model of the NFT rewards system is undermined since tiers are supposed to represent specific contribution levels
- Protocol can mint more NFTs than intended, diluting value for original tier holders

Likelihood: High

- ETH volatility is common and can easily cross the $50 threshold difference
- The protocol explicitly markets these as tier-based rewards with fixed USD thresholds
- All users are affected equally by this issue

Proof of Concept:

1. User A funds 0.02 ETH when ETH/USD = $2,500 (value = $50 USD)
2. User A waits, ETH price rises to ETH/USD = $3,000
3. User A claims NFT - their $50 ETH is now worth $60 USD
4. User A legitimately receives Bronze NFT

Adversarial Scenario:

1. User B funds 0.016 ETH when ETH/USD = $3,125 (value = $49.99 USD, below threshold)
2. ETH price drops to ETH/USD = $2,500
3. User B's $49.99 is now worth ~$40 USD
4. User B cannot claim Bronze NFT

Scenario where it works against protocol:

1. ETH price rises 20% after funding period
2. Many users who funded just below thresholds now qualify for higher tiers
3. More NFTs minted than intended, diluting value for original tier holders

PoC Test (Foundry)

```
/// @title CR-1: Demonstrate tier dilution when price increases
/// @notice Users who funded just below threshold at low price get NFTs when price increases
function test_CR1_TierDilution_PriceIncreaseAfterFunding() public _openCampaignDefault {
    // Setup: ETH price = $1923 (just below threshold for 0.026 ETH to be $50)
    mockPriceFeed.updateAnswer(int256(1923e8));

    // User funds 0.0259 ETH = ~$49.80 (below $50 threshold)
    vm.startPrank(funder);
    fundMe.fund{value: 0.0259 ether}();
    vm.stopPrank();

    uint256 roundCampaign = fundMe.getCurrentRound();

    // Verify user cannot claim
    vm.prank(funder);
    vm.expectRevert();
    fundMe.claimRewardNft(roundCampaign);

    // Now price increases to $3000
    mockPriceFeed.updateAnswer(int256(3000e8));

    // Now user's 0.0259 ETH is worth: 0.0259 * 3000 = $77.70 > $50
    // User should now be able to claim Bronze
    vm.prank(funder);
    fundMe.claimRewardNft(roundCampaign);

    bool hasMinted = fundMe.getHasFunderMintedTier(roundCampaign, funder, NftTier.BRONZE);
    assertTrue(hasMinted, "User funded below threshold but claimed NFT due to price increase");

    console.log("VULNERABILITY CONFIRMED: User received NFT despite funding below threshold");
}
```

Recommended Mitigation:

Store the USD amount at fund time:

```
// Add new mapping to track USD amounts at fund time
+ mapping(uint256 currentRound => mapping(address funder => uint256 usdAmount)) private s_roundToFunderUsdAmount;
```

```

function fund() public payable {
    uint256 currentRound = s_currentRound;

    // Check FINALIZED state
    if (s_roundToCampignStatus[currentRound] != CampaignLifeCycle.OPEN) {
        revert FundMe__CampaignStatusMustBeOpen();
    }

    uint256 usdAmount = msg.value.getConversionRate(s_priceFeed);

    if (usdAmount < MINIMUM_USD) {
        revert FundMe__NotEnoughFunds();
    }

    // Store USD amount at fund time
+   s_roundToFunderUsdAmount[currentRound][msg.sender] += usdAmount;

    if (s_roundToAddressToAmountFunder[currentRound][msg.sender] == 0) {
        s_roundToFunders[currentRound].push(msg.sender);
    }

    s_roundToAddressToAmountFunder[currentRound][msg.sender] += msg.value;
    s_roundToContractBalance[currentRound] += msg.value;

    emit Funded(currentRound, msg.sender, msg.value, usdAmount);
}

function claimRewardNft(uint256 roundCampaign) public {
    // ...

-   uint256 totalFundedEth = s_roundToAddressToAmountFunder[roundCampaign][msg.sender];
-   uint256 totalFundedUsd = totalFundedEth.getConversionRate(s_priceFeed);
+   uint256 totalFundedUsd = s_roundToFunderUsdAmount[roundCampaign][msg.sender];

    if (totalFundedUsd < BRONZE_THRESHOLD) {
        revert FundMe__BalanceHaveToAtLeastEqualBronzeThreshold();
    }
    // ...
}
```

Medium

[M-1] No Minimum Campaign Duration Enforced

Description:

The `openCampaign` function only checks that `_deadlineTimeStamp > block.timestamp`, allowing campaigns with extremely short durations (e.g., 1 second). While this may be intentional for testing, it could be exploited by a malicious owner to create and immediately close campaigns, potentially griefing funders who send ETH expecting a longer campaign window.

```
function openCampaign(uint256 _fundraisingTarget, uint256 _deadlineTimeStamp) public onlyOwner {
    if (_fundraisingTarget == 0) {
        revert FundMe__FundraisingTargetMustBeMoreThanZero();
    }
    if (_deadlineTimeStamp <= block.timestamp) {
        revert FundMe__DeadlineTimeStampMustBeBiggerThanNow();
    }
    // @> Missing: minimum duration check
    // ...
}
```

Impact: Medium

- Funders could be griefed into sending ETH to campaigns that close immediately
- Gas loss from failed transactions if campaign closes before confirmation
- Creates misleading UX where campaign appears open but closes instantly

Likelihood: Medium

- Requires malicious owner intent (or testing scenarios)
- Testing scenarios frequently use 0-duration campaigns
- Legitimate use cases would have proper durations

Proof of Concept:

PoC Test (Foundry)

```
/// @title M-1: Campaign can be opened with 1 second duration
/// @notice This test demonstrates that campaigns with extremely short durations are allowed
function test_M1_CampaignCanOpenWithOneSecondDuration() public {
    // When: Owner opens campaign with 1 second duration
    vm.prank(owner);
    // This should NOT revert - demonstrating the vulnerability
    fundMe.openCampaign(FUNDRAISING_TARGET, block.timestamp + 1 seconds);
```

```

    // Then: Campaign is OPEN
    uint256 currentRound = fundMe.getCurrentRound();
    assertEq(uint256(fundMe.getRoundToCampignStatus(currentRound)),
        uint256(CampaignLifeCycle.OPEN));

}
```

Recommended Mitigation:

Add minimum campaign duration check:

```
+uint256 private constant MINIMUM_CAMPAIGN_DURATION = 1 hours;

function openCampaign(uint256 _fundraisingTarget, uint256
    _deadlineTimeStamp) public onlyOwner {
    if (_fundraisingTarget == 0) {
        revert FundMe__FundraisingTargetMustBeMoreThanZero();
    }
    if (_deadlineTimeStamp <= block.timestamp) {
        revert FundMe__DeadlineTimeStampMustBeBiggerThanNow();
    }
+   if (_deadlineTimeStamp - block.timestamp <
    MINIMUM_CAMPAIGN_DURATION) {
+       revert FundMe__CampaignDurationTooShort();
+   }
    // ...
}
```

Low

[L-1] Owner Address Not Validated in Constructor

Description:

`FundMe` constructor sets `i_owner` from `msg.sender` without validating against the zero address. If `msg.sender` is `address(0)` (which can happen during contract creation via `CREATE` or certain factory patterns), the contract will be permanently broken with no upgrade path since `i_owner` is immutable.

```
constructor(AggregatorV3Interface priceFeed) {
    if (address(priceFeed) == address(0)) {
        revert FundMe__ZeroAddressNotAllowed();
    }

```

```
    i_owner = msg.sender; // @> No zero address check
    s_priceFeed = AggregatorV3Interface(priceFeed);
    s_fundMeRewardNft = new FundMeRewardNft(address(this));
    s_roundToCampignStatus[s_currentRound] = CampaignLifeCycle.
        NOT_OPEN_YET;
}
```

Impact: High

- Contract permanently broken if deployed with `address(0)` as deployer
- All funds locked with no recovery
- All `onlyOwner` functions unusable

Likelihood: Low

- Unlikely in normal deployments
- Possible in factory/fork scenarios
- Standard practice is to trust `msg.sender`

Recommended Mitigation:

Add zero address validation:

```
constructor(AggregatorV3Interface priceFeed) {
    if (address(priceFeed) == address(0)) {
        revert FundMe__ZeroAddressNotAllowed();
    }

+   if (msg.sender == address(0)) {
+       revert FundMe__ZeroAddressNotAllowed();
+   }
    i_owner = msg.sender;
    s_priceFeed = AggregatorV3Interface(priceFeed);
    s_fundMeRewardNft = new FundMeRewardNft(address(this));
}
```

[L-2] `closeCampaign` Has Redundant Storage Reads

Description:

`closeCampaign` checks `roundCampaign != s_currentRound` but then reads `s_currentRound` again from storage for the status check. Since we've already validated they are equal, we should use the local `roundCampaign` variable.

```
function closeCampaign(uint256 roundCampaign) public {
    if (roundCampaign != s_currentRound) {
        revert FundMe__InvalidRoundCampaign();
    }

    if (s_roundToCampignStatus[s_currentRound] != CampaignLifeCycle.
        OPEN) { // @>
        revert FundMe__CampaignStatusMustBeOpen();
    }
    // ...
}
```

Impact: Low

- Gas waste from extra SLOAD
- Does not affect security, only runtime cost

Likelihood: High

- Occurs on every `closeCampaign` call
- This is a gas optimization only

Recommended Mitigation:

Use the local variable after validation:

```
function closeCampaign(uint256 roundCampaign) public {
    if (roundCampaign != s_currentRound) {
        revert FundMe__InvalidRoundCampaign();
    }

-   if (s_roundToCampignStatus[s_currentRound] != CampaignLifeCycle.
    OPEN) {
+   if (s_roundToCampignStatus[roundCampaign] != CampaignLifeCycle.OPEN
    ) {
        revert FundMe__CampaignStatusMustBeOpen();
    }
    // Use roundCampaign instead of s_currentRound for subsequent reads
    // ...
}
```

[L-3] Single Point of Failure - Owner Controls All Critical Operations

Description:

All critical functions (`openCampaign`, `closeCampaign`, `withdraw`) are restricted to the contract owner. If the owner's private key is compromised, an attacker can:

1. Open campaigns with malicious parameters
2. Withdraw all ETH from any closed campaign
3. Prevent legitimate campaign operations

While this is by design for a centralized crowdfunding platform, it creates significant counterparty risk for funders.

```
function openCampaign(...) public onlyOwner { ... }
function withdraw(uint256 roundCampaign) public onlyOwner nonReentrant
    { ... }
```

Impact: High

- Complete protocol compromise if owner key is leaked
- No on-chain recovery mechanism exists
- All user funds at risk

Likelihood: Low

- Depends on owner's security practices
- By design for centralized platform
- Standard risk for single-owner contracts

Recommended Mitigation:

Consider adding:

1. **Time-lock for critical actions:**

```
uint256 public constant OWNER_ACTION_DELAY = 2 days;

mapping(bytes32 => uint256) public pendingOwnerActions;

function initiateOwnerAction(bytes32 actionHash) external onlyOwner {
    pendingOwnerActions[actionHash] = block.timestamp +
        OWNER_ACTION_DELAY;
}

function executeOwnerAction(bytes32 actionHash) external {
    require(pendingOwnerActions[actionHash] != 0, "Action not pending")
        ;
    require(block.timestamp >= pendingOwnerActions[actionHash], "Too
        early");
    pendingOwnerActions[actionHash] = 0;
    // Execute the action
```

```
}
```

2. **Multi-signature support:**

```
mapping(address => bool) public isAdmin;
uint256 public constant MIN_APPROVALS = 2;

mapping(bytes32 => mapping(address => bool)) public approvedActions;

function approveAction(bytes32 actionHash) external {
    require(isAdmin[msg.sender], "Not admin");
    approvedActions[actionHash][msg.sender] = true;
}

function executeAction(bytes32 actionHash) external {
    uint256 approvalCount = 0;
    for (/* all admins */) {
        if (approvedActions[actionHash][admin]) approvalCount++;
    }
    require(approvalCount >= MIN_APPROVALS, "Not enough approvals");
    // Execute
}
```

[L-4] Redundant Storage Write in Constructor Wastes Gas on Every Deployment

Description:

The constructor explicitly sets `s_roundToCampignStatus[s_currentRound] = CampaignLifeCycle.NOT_OPEN_YET`, but since `s_currentRound` is initialized to 0 and `CampaignLifeCycle.NOT_OPEN_YET` is the first enum value (also 0), this write is completely redundant.

```
uint256 private s_currentRound; // defaults to 0

constructor(AggregatorV3Interface priceFeed) {
    // ...
    s_roundToCampignStatus[s_currentRound] = CampaignLifeCycle.
        NOT_OPEN_YET; // redundant
}
```

Impact: Low

- Wastes approximately 20,000-50,000 gas per deployment
- Does not affect security, only deployment cost

Likelihood: High

- Affects every single deployment

Recommended Mitigation:

Remove the redundant initialization:

```
constructor(AggregatorV3Interface priceFeed) {
    if (address(priceFeed) == address(0)) {
        revert FundMe__ZeroAddressNotAllowed();
    }

    i_owner = msg.sender;
    s_priceFeed = AggregatorV3Interface(priceFeed);
    s_fundMeRewardNft__ERC721 = new FundMeRewardNft(address(this));
-   s_roundToCampignStatus[s_currentRound] = CampaignLifeCycle.
    NOT_OPEN_YET;
}
```

[L-5] Solidity Version Uses Caret Operator Instead of Exact Version

Description:

All contracts use `pragma solidity ^0.8.19;` which allows compilation with any Solidity version from 0.8.19 up to (but not including) 0.9.0.

```
// All contracts in ./src/
pragma solidity ^0.8.19; // Should be: pragma solidity 0.8.19;
```

Impact: Low

- Bytecode may differ between compilation environments
- Future compiler versions could introduce unexpected behavior

Likelihood: Medium

- Common practice, but not best practice for security-critical contracts

Recommended Mitigation:

Pin to exact compiler version:

```
-pragma solidity ^0.8.19;
+pragma solidity 0.8.19;
```

[L-6] Ambiguous Ownership - NFT URI Can Only Be Changed by FundMe Contract, Not FundMe Owner

Description:

The `FundMeRewardNft.setTierUri()` function has `onlyOwner` modifier, suggesting the owner should be able to modify NFT tier URIs. However, the ownership chain is ambiguous:

- `FundMeRewardNft` is owned by the FundMe contract (not FundMe's owner)
- FundMe owner cannot directly call `setTierUri()` because they are not the owner
- If ownership is changed to FundMe's owner, then FundMe contract cannot mint NFTs anymore

This creates ambiguity in the design intent:

```
// FundMeRewardNft.sol:20 - Owned by FundMe contract
constructor(address fundMe_) ERC721("FundMe Reward Nft", "FRN") Ownable
    (fundMe_) {}
```

```
s_fundMeRewardNft__ERC721 = new FundMeRewardNft(address(this));
```

Impact: Low

- FundMe owner cannot modify NFT tier URIs directly
- No owner-controlled way to update metadata exists
- Could be intentional design (ownership by contract) but unclear

Likelihood: Low

- This appears to be by design (FundMe contract owns NFT for minting)
- Unclear if owner should be able to change URIs

Recommended Mitigation:

Add a "drill-through" function in FundMe that allows owner to modify NFT URIs:

```
function setNftTierUri(NftTier tier, string memory uri) external
    onlyOwner {
     s_fundMeRewardNft__ERC721.setTierUri(tier, uri);
}
```

This allows FundMe owner to change NFT URIs while keeping FundMe contract as the technical owner for minting purposes.

Proof of Concept:

PoC Test (Foundry)

```
/// @title H-1: FundMe owner cannot change tier URIs
/// @notice Demonstrates that the ownership chain is broken
function test_H1_FundMeOwnerCannotChangeUri() public {
    // Mint a Bronze NFT
    vm.startPrank(address(fundMe));
    uint256 tokenId = fundMeRewardNft.mintNft(funder, NftTier.BRONZE);
    vm.stopPrank();

    // Try to change URI as FundMe owner (not FundMe contract)
    // This will REVERT because owner is FundMe contract, not FundMe's owner
    string memory newUri = "ipfs://owner-want-to-change-uri";
    vm.prank(owner);
    vm.expectRevert();
    fundMeRewardNft.setTierUri(NftTier.BRONZE, newUri);

    console.log("ISSUE: FundMe owner cannot modify NFT URIs");
    console.log("DESIGN PROBLEM: Ownership chain is broken");
}

/// @title H-1: NFT metadata can still be changed by FundMe contract
function test_H1_NFTContractCanChangeUri() public {
    // Mint a Bronze NFT
    vm.startPrank(address(fundMe));
    uint256 tokenId = fundMeRewardNft.mintNft(funder, NftTier.BRONZE);
    vm.stopPrank();

    // Get original URI
    string memory originalUri = fundMeRewardNft.tokenURI(tokenId);
    console.log("Original URI:", originalUri);

    // FundMe contract (owner of NFT) changes URI
    string memory newUri = "ipfs://new-uri";
    vm.prank(address(fundMe));
    fundMeRewardNft.setTierUri(NftTier.BRONZE, newUri);

    // Verify changed
    assertEq(fundMeRewardNft.tokenURI(tokenId), newUri);
    console.log("URI was changed by FundMe contract");
}
```

Recommended Mitigation:

Fix the ownership chain by adding a function in FundMe that allows the owner to modify NFT URIs:

```
+function setNftTierUri(NftTier tier, string memory uri) external onlyOwner {
+    s_fundMeRewardNft.setTierUri(tier, uri);
+}

// FundMeRewardNft.sol - Add a way to freeze URIs
```

```
+mapping(uint256 tokenId => string originalUri) private s_tokenIdToOriginalUri;

function mintNft(address to, NftTier tier) public onlyOwner returns (uint256) {
    uint256 tokenId = s_tokenCounter;
    s_tokenIdToTier[s_tokenCounter] = tier;
+   s_tokenIdToOriginalUri[s_tokenCounter] = s_tierToTokenURI[tier]; // Snapshot
    _safeMint(to, s_tokenCounter);
    s_tokenCounter++;
    return tokenId;
}
```

Informational

[I-1] Comment Syntax Inconsistency with Audit Tags

Description:

The codebase uses inconsistent comment styles for audit-related annotations. Some use `// ? @audit-` prefix while others use `// @audit-` or `//! @>`. This inconsistency makes it harder to grep for all audit-related comments.

```
// ? @audit-info address 0 check on msg.sender
// ? @audit-gas cache s_currentRound in memory...
// ? @audit-info improve GPT ahh comment
// ? @audit-info external function for saving more gas
// ? @audit-info reentrancy posible with onERC721Received...
```

Recommended Mitigation:

Standardize audit comment format:

```
// @audit-info: <info message>
// @audit-gas: <gas optimization suggestion>
// @audit-warning: <potential issue, needs review>
// @audit-severity: <severity if confirmed>
```

[I-2] Redundant Comment "Check if first time funding"

Description:

The comment `// <icon> GAS OPTIMIZATION: Check if first time funding` at line 164 is self-explanatory and the code is clear enough without it. Comments that simply restate what the code does add noise without providing additional value.

```
// ? @audit-info improve GPT ahh comment
// <icon> GAS OPTIMIZATION: Check if first time funding
if (s_roundToAddressToAmountFunder[currentRound][msg.sender] == 0) {
    s_roundToFunders[currentRound].push(msg.sender);
}
```

Recommended Mitigation:

Remove redundant comments or improve them to explain *why* rather than *what*:

```
// Only add sender to funders array once to prevent duplicates
if (s_roundToAddressToAmountFunder[currentRound][msg.sender] == 0) {
    s_roundToFunders[currentRound].push(msg.sender);
}
```

[I-3] Race Condition Between `closeCampaign` and `fund` Functions

Description:

The `closeCampaign` function is publicly callable with no access control (as documented in README). While it checks that the campaign status is `OPEN`, there's a potential front-running race condition: an attacker could observe a pending `fund()` transaction and front-run with `closeCampaign()` to prevent the funder from participating, if the conditions happen to align at that moment.

```
function closeCampaign(uint256 roundCampaign) public {
    if (roundCampaign != s_currentRound) {
        revert FundMe__InvalidRoundCampaign();
    }
    if (s_roundToCampignStatus[s_currentRound] != CampaignLifeCycle.
        OPEN) {
        revert FundMe__CampaignStatusMustBeOpen();
    }
    // ...
}
```

Impact: Low

- A funder's transaction could be blocked if conditions align (deadline just passed, target reached)
- Griefing attack requires specific timing but is possible
- Disrupts user experience and can cause missed funding opportunities

Likelihood: Low

- Requires attacker to monitor mempool and react within same block
- MEV bots commonly perform such attacks
- Window is narrow but real
- By design per README — intentionally public

Proof of Concept:

PoC Test (Foundry)

```
/// @title I-3: Front-running fund() with closeCampaign
/// @notice Demonstrates MEV-style front-running attack
function test_I3_FrontRunFundWithCloseCampaign() public _openCampaignDefault {
    // Scenario: Funder tries to fund, attacker front-runs with closeCampaign

    uint256 currentRound = fundMe.getCurrentRound();

    // Warp to right before deadline
    vm.warp(block.timestamp + DEADLINE_DURATION - 1);

    // Attacker sees pending fund() and front-runs with closeCampaign
    // In real scenario, this would be detected from mempool
    vm.prank(attacker);
    fundMe.closeCampaign(currentRound);

    // Now funder's fund() will fail
    vm.startPrank(funder);
    vm.expectRevert();
    fundMe.fund{value: 1 ether}();
    vm.stopPrank();

    // Verify campaign is closed
    assertEq(uint256(fundMe.getRoundToCampignStatus(currentRound)), uint256(CampaignLifeCycle.CLOSED));

    console.log("ISSUE CONFIRMED: Fund transaction can be blocked");
}
```

Recommended Mitigation:

No practical mitigation available without changing the permissionless design. The `closeCampaign` function is intentionally public (per README: "a manual call to `closeCampaign()` by users/owner"). The race condition is inherent to this design choice and is classified as informational.

If stricter control is desired:

```
-function closeCampaign(uint256 roundCampaign) public {
```

```
+function closeCampaign(uint256 roundCampaign) public onlyOwner {
    // ...
}
```

Appendix: Additional Recommendations

A. Consider Adding ReentrancyGuard to `claimRewardNft`

While the current implementation has `nonReentrant` on `withdraw()`, the `claimRewardNft` function mints NFTs via `_safeMint`, which could trigger `onERC721Received` callbacks. Consider adding `nonReentrant` for consistency:

```
function claimRewardNft(uint256 roundCampaign) public nonReentrant {
    // ...
}
```

B. Consider Emitting Events for Tier Claims

Currently, `RewardNftMinted` is emitted but could include more context:

```
event RewardNftMinted(
    address indexed to,
    uint256 indexed tokenId,
    NftTier tier,
    uint256 roundCampaign,
    uint256 usdAmount
);
```

C. Gas Optimization: Cache Storage Reads

Multiple functions read `s_currentRound` from storage. Consider caching in memory when used multiple times within a function.

FundMeAuditTest.t.sol

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.19;

import {Test, console} from "forge-std/Test.sol";
import {FundMe} from "src/FundMe.sol";
import {FundMeRewardNft} from "src/FundMeRewardNft.sol";
import {DeployFundMe} from "script/DeployFundMe.s.sol";
import {PriceConverter} from "src/PriceConverter.sol";
import {NftTier} from "src/NftTier.sol";
import {CampaignLifeCycle} from "src/CampaignLifeCycle.sol";
import {CampaignClosedReason} from "src/CampaignClosedReason.sol";
import {AggregatorV3Interface} from "@chainlink/contracts/src/v0.8/
    shared/interfaces/AggregatorV3Interface.sol";
import {MockV3Aggregator} from "@chainlink/contracts/src/v0.8/tests/
    MockV3Aggregator.sol";

abstract contract CodeConstant {
    uint256 public constant SEND_MINIMUM_ETH = 0.0025 ether;
    uint256 public constant SEND_UNDER_MINIMUM_ETH = 0.0014 ether;
    uint256 public constant RICH_FUNDER = 1 ether;

    // Thrashold for each tier
    uint256 public constant BRONZE_THRESHOLD = 0.026 ether;
    uint256 public constant SILVER_THRESHOLD = 0.1 ether;
    uint256 public constant GOLD_THRESHOLD = 0.5 ether;

    // Campaign configuration
    uint256 public FUNDRAISING_TARGET = 10000e18;
    uint256 public constant DEADLINE_DURATION = 1 hours;
    uint256 public DEADLINE_TIMESTAMP = block.timestamp +
        DEADLINE_DURATION;
}

// Kontrak Mock untuk memaksa kegagalan transfer ETH
contract Rejector {
    receive() external payable {
        revert("I refuse any ETH");
    }
}

contract FundMeAuditTest is CodeConstant, Test {
    using PriceConverter for uint256;

    /* State Variables */
    FundMe public fundMe;
    FundMeRewardNft public fundMeRewardNft;
    DeployFundMe public deployer;
    MockV3Aggregator public mockPriceFeed;
```

```

    address public owner = makeAddr("owner");
    address public funder = makeAddr("funder");
    address public attacker = makeAddr("attacker");

    int256 public constant ETH_PRICE_HIGH = 3000e8; // $3000/ETH (high price scenario)
    int256 public constant ETH_PRICE_LOW = 2000e8;  // $2000/ETH (low price scenario)
    int256 public constant ETH_PRICE_MID = 2500e8;  // $2500/ETH (mid price)

    event CampaignClosed(uint256 indexed roundCampaign, CampaignClosedReason indexed reason);
    event RewardNftMinted(address indexed to, uint256 tokenId, NftTier tier);

    modifier _openCampaign(uint256 duration) {
        vm.prank(owner);
        fundMe.openCampaign(FUNDRAISING_TARGET, block.timestamp + duration);
        _;
    }

    modifier _openCampaignDefault() {
        vm.prank(owner);
        fundMe.openCampaign(FUNDRAISING_TARGET, block.timestamp + DEADLINE_DURATION);
        _;
    }

    function setUp() public {
        // Deploy mock price feed with initial price
        mockPriceFeed = new MockV3Aggregator(8, ETH_PRICE_MID);

        // Deploy FundMe with mock price feed
        vm.prank(owner);
        fundMe = new FundMe(AggregatorV3Interface(address(mockPriceFeed)));
        fundMeRewardNft = fundMe.getFundMeRewardNft();

        // Fund test accounts
        vm.deal(funder, 100e18);
        vm.deal(attacker, 100e18);
    }

    // ============================================================
    // [CR-1] NFT Tier Eligibility Calculated at Claim Time, Not Fund Time
    // ============================================================

```

```
    // CR-1: Demonstrate tier dilution when price increases
    // Users who funded just below threshold at low price get NFTs when price increases
    // This test CONFIRMS the vulnerability exists
    function test_CR1_TierDilution_PriceIncreaseAfterFunding() public _openCampaignDefault {
        // Setup: ETH price = $1923 (just below threshold for 0.026 ETH to be $50)
        mockPriceFeed.updateAnswer(int256(1923e8));

        // User funds 0.0259 ETH = ~$49.80 (below $50 threshold)
        vm.startPrank(funder);
        fundMe.fund{value: 0.0259 ether}();
        vm.stopPrank();

        uint256 roundCampaign = fundMe.getCurrentRound();

        // Verify user cannot claim
        vm.prank(funder);
        vm.expectRevert();
        fundMe.claimRewardNft(roundCampaign);

        // Now price increases to $3000
        mockPriceFeed.updateAnswer(int256(ETH_PRICE_HIGH));

        // Now user's 0.0259 ETH is worth: 0.0259 * 3000 = $77.70 > $50
        // User should now be able to claim Bronze
        vm.prank(funder);
        fundMe.claimRewardNft(roundCampaign);

        bool hasMinted = fundMe.getHasFunderMintedTier(roundCampaign, funder, NftTier.BRONZE);
        assertTrue(hasMinted, "User funded below threshold but claimed NFT due to price increase");

        console.log("VULNERABILITY CONFIRMED: User received NFT despite funding below threshold");
    }

    // CR-1: User loses tier eligibility when price drops
    function test_CR1_UserLosesEligibilityWhenPriceDrops() public _openCampaignDefault {
        // Given: ETH price is $3000, user funds enough for Bronze
        mockPriceFeed.updateAnswer(int256(ETH_PRICE_HIGH)); // $3000/ETH

        vm.startPrank(funder);
        fundMe.fund{value: 0.02 ether}(); // 0.02 ETH = $60 at $3000
        vm.stopPrank();

        uint256 roundCampaign = fundMe.getCurrentRound();
```

```

        // User can claim Bronze
        vm.prank(funder);
        fundMe.claimRewardNft(roundCampaign);

        bool hasMinted = fundMe.getHasFunderMintedTier(roundCampaign, funder, NftTier.BRONZE);
        assertTrue(hasMinted, "User should have minted Bronze at $3000/ETH");

        // Now price drops to $1500 - user already claimed so no issue
        mockPriceFeed.updateAnswer(int256(1500e8));

        console.log("INFO: User already claimed NFT before price drop");
        console.log("ISSUE: If price dropped BEFORE claiming, user would lose eligibility");
    }

    // ============================================================
    // [L-6] Ambiguous Ownership - NFT URI Can Only Be Changed by FundMe Contract
    // ============================================================

    // L-6: Owner can change tier URI after NFT is minted
    // This test demonstrates that NFT metadata is not frozen after minting
    // NOTE: FundMeRewardNft's owner is the FundMe contract itself
    // This is a design issue - the FundMe owner cannot modify NFT URIs
    function test_L6_OwnerCanChangeTierUriAfterMint() public {
        // Given: FundMeRewardNft is owned by FundMe contract
        // The FundMe contract itself is the owner of the NFT contract
        address nftOwner = fundMeRewardNft.owner();
        assertEq(nftOwner, address(fundMe), "NFT contract owner should be FundMe");

        // Mint a Bronze NFT (FundMe can mint because it's the owner)
        vm.startPrank(address(fundMe));
        uint256 tokenId = fundMeRewardNft.mintNft(funder, NftTier.BRONZE);
        vm.stopPrank();

        // Get original URI
        string memory originalUri = fundMeRewardNft.tokenURI(tokenId);
        console.log("Original URI:", originalUri);

        // When: FundMe contract changes the tier URI (via direct call)
        string memory newUri = "ipfs://new-metadata-uri-that-changes-everything";
        vm.prank(address(fundMe));
        fundMeRewardNft.setTierUri(NftTier.BRONZE, newUri);
```

```

        // Then: The minted NFT now shows the NEW URI
        string memory updatedUri = fundMeRewardNft.tokenURI(tokenId);
        console.log("Updated URI:", updatedUri);

        // This passes, demonstrating the issue
        assertEq(updatedUri, newUri, "URI was changed after minting");
        assertFalse(keccak256(bytes(originalUri)) == keccak256(bytes(newUri)), "URI content changed");

        console.log("ISSUE: NFT metadata can be changed after minting");
        console.log("BUT: FundMe owner cannot change URIs directly (FundMe contract is NFT owner)");
    }

    // L-6: FundMe owner cannot change tier URIs
    // This demonstrates that the ownership chain is broken
    function test_L6_FundMeOwnerCannotChangeUri() public {
        // Mint a Bronze NFT
        vm.startPrank(address(fundMe));
        fundMeRewardNft.mintNft(funder, NftTier.BRONZE);
        vm.stopPrank();

        // Try to change URI as FundMe owner (not FundMe contract)
        string memory newUri = "ipfs://hacked-uri";
        vm.prank(owner);
        vm.expectRevert();
        fundMeRewardNft.setTierUri(NftTier.BRONZE, newUri);

        console.log("ISSUE: FundMe owner cannot modify NFT URIs");
        console.log("DESIGN PROBLEM: Ownership chain is broken");
    }

    // ============================================================
    // [L-4] Redundant Storage Write in Constructor
    // ============================================================

    // L-4: Document gas optimization issue
    function test_L4_DocumentGasWaste() public view {
        console.log("NOTE: L-4 is a gas optimization issue");
        console.log("s_roundToCampignStatus[0] is set to NOT_OPEN_YET (0)");
        console.log("But this is the default value, so it's redundant");

        // Show the enum value
        assertEq(uint256(CampaignLifeCycle.NOT_OPEN_YET), 0);
        assertEq(uint256(fundMe.getRoundToCampignStatus(0)), uint256(CampaignLifeCycle.NOT_OPEN_YET));

```

```
        console.log("RECOMMENDATION: Remove redundant SSTORE in constructor");
    }

    // ============================================================
    // [M-1] No Minimum Campaign Duration Enforced
    // ============================================================

    // M-1: Campaign can be opened with 1 second duration
    // This test demonstrates that campaigns with extremely short durations are allowed
    // This test CONFIRMS the vulnerability exists
    function test_M1_CampaignCanOpenWithOneSecondDuration() public {
        // When: Owner opens campaign with 1 second duration
        vm.prank(owner);
        // This should NOT revert - demonstrating the vulnerability
        fundMe.openCampaign(FUNDRAISING_TARGET, block.timestamp + 1 seconds);

        // Then: Campaign is OPEN
        uint256 currentRound = fundMe.getCurrentRound();
        assertEq(uint256(fundMe.getRoundToCampignStatus(currentRound)), uint256(CampaignLifeCycle.OPEN));

        console.log("VULNERABILITY CONFIRMED: Campaign opened with 1 second duration");
    }

    // M-1: Attack scenario - immediate griefing
    // Demonstrates how owner can grief funders with instant campaign
    function test_M1_GriefingWithInstantCampaign() public _openCampaignDefault {
        // Given: Campaign is open, but we need to warp to deadline first
        vm.warp(block.timestamp + DEADLINE_DURATION + 1);

        // When: Owner closes it
        vm.prank(owner);
        fundMe.closeCampaign(fundMe.getCurrentRound());

        // Then: Campaign is CLOSED
        uint256 currentRound = fundMe.getCurrentRound();
        assertEq(uint256(fundMe.getRoundToCampignStatus(currentRound)), uint256(CampaignLifeCycle.CLOSED));

        // And: Funder cannot fund anymore
        vm.startPrank(funder);
        vm.expectRevert();
        fundMe.fund{value: 1 ether}();
        vm.stopPrank();

```

```
        console.log("VULNERABILITY CONFIRMED: Instant campaigns enable griefing");
    }

    // ============================================================
    // [I-3] Race Condition Between closeCampaign and fund
    // ============================================================


    // I-3: Front-running fund() with closeCampaign
    // Demonstrates MEV-style front-running attack (by design per README)
    function test_I3_FrontRunFundWithCloseCampaign() public _openCampaignDefault {
        // Scenario: Funder tries to fund, attacker front-runs with closeCampaign

        uint256 currentRound = fundMe.getCurrentRound();

        // Warp to after deadline
        vm.warp(block.timestamp + DEADLINE_DURATION + 1);

        // Attacker sees pending fund() and front-runs with closeCampaign
        // In real scenario, this would be detected from mempool
        vm.prank(attacker);
        fundMe.closeCampaign(currentRound);

        // Now funder's fund() will fail
        vm.startPrank(funder);
        vm.expectRevert();
        fundMe.fund{value: 1 ether}();
        vm.stopPrank();

        // Verify campaign is closed
        assertEq(uint256(fundMe.getRoundToCampignStatus(currentRound)), uint256(CampaignLifeCycle.CLOSED));

        console.log("ISSUE CONFIRMED: Fund transaction can be blocked (by design)");
    }

    // I-3: Multiple users can race to close
    function test_I3_MultipleUsersCanRaceToClose() public _openCampaignDefault {
        address user1 = makeAddr("user1");
        address user2 = makeAddr("user2");
        vm.deal(user1, 10e18);
        vm.deal(user2, 10e18);

        uint256 currentRound = fundMe.getCurrentRound();
```

```

        // Warp to after deadline
        vm.warp(block.timestamp + DEADLINE_DURATION + 1);

        // Both users can close the campaign
        vm.prank(user1);
        fundMe.closeCampaign(currentRound);

        // Verify it's closed
        assertEq(uint256(fundMe.getRoundToCampignStatus(currentRound)), uint256(CampaignLifeCycle.CLOSED));

        // User2's call will fail with different error (not access control)
        vm.prank(user2);
        vm.expectRevert();
        fundMe.closeCampaign(currentRound);

        console.log("ISSUE CONFIRMED: No access control on closeCampaign (by design)");
    }

    // ============================================================
    // [L-1] Owner Address Not Validated in Constructor
    // ============================================================

    // L-1: Document address(0) limitation
    // Testing address(0) as deployer is difficult without factory pattern
    function test_L1_DocumentAddressZeroLimitation() public view {
        console.log("LIMITATION: Cannot test address(0) deployer in Foundry without factory");
        console.log("RECOMMENDATION: Add validation in constructor");
        assertTrue(true);
    }

    // L-1: Show impact of address(0) owner
    function test_L1_ShowImpactOfAddressZeroOwner() public {
        // Show that current implementation trusts msg.sender
        address testDeployer = address(this);
        vm.prank(testDeployer);
        FundMe testFundMe = new FundMe(AggregatorV3Interface(address(mockPriceFeed)));

        // Owner is set to deployer
        assertEq(testFundMe.getOwner(), testDeployer);

        console.log("NOTE: No validation prevents address(0) as owner");
    }

```

```
    // ============================================================
    // [L-2] Redundant Storage Reads - Not directly testable
    // ============================================================

    // L-2: Document gas optimization issue
    // Gas issues can be verified with gas snapshots
    function test_L2_DocumentGasIssue() public view {
        console.log("NOTE: L-2 is a gas optimization, not a security issue");
        console.log("RECOMMENDATION: Use local variable after equality check");
        console.log("VERIFICATION: Use `forge snapshot` to compare gas before/after fix");
        assertTrue(true);
    }

    // ============================================================
    // [L-3] Single Point of Failure - Design Issue
    // ============================================================

    // L-3: Demonstrate owner can withdraw all funds
    // Shows the centralization risk
    function test_L3_OwnerCanWithdrawAllFunds() public _openCampaignDefault {
        // Given: Funder sends ETH
        vm.startPrank(funder);
        fundMe.fund{value: 10 ether}();
        vm.stopPrank();

        uint256 currentRound = fundMe.getCurrentRound();
        uint256 contractBalance = address(fundMe).balance;
        console.log("Contract balance:", contractBalance);

        // Warp to deadline and close
        vm.warp(block.timestamp + DEADLINE_DURATION + 1);

        // Close by anyone (documenting that closeCampaign is permissionless)
        fundMe.closeCampaign(currentRound);

        // When: Owner withdraws
        uint256 ownerBalanceBefore = owner.balance;
        vm.prank(owner);
        fundMe.withdraw(currentRound);
        uint256 ownerBalanceAfter = owner.balance;

        // Then: All ETH goes to owner
        assertEq(ownerBalanceAfter - ownerBalanceBefore, contractBalance);
        assertEq(address(fundMe).balance, 0);

```

```
        console.log("CENTRALIZATION RISK: All funds withdrawn by single owner");
    }

    // ============================================================
    // Helper Functions for Debugging
    // ============================================================

    function test_Debug_ShowThresholds() public view {
        console.log("=== Threshold Values ===");
        console.log("BRONZE_THRESHOLD:", BRONZE_THRESHOLD);
        console.log("SILVER_THRESHOLD:", SILVER_THRESHOLD);
        console.log("GOLD_THRESHOLD:", GOLD_THRESHOLD);

        console.log("\n=== At $2500/ETH ===");
        console.log("BRONZE USD value:", BRONZE_THRESHOLD * 2500 / 1e18);
        console.log("SILVER USD value:", SILVER_THRESHOLD * 2500 / 1e18);
        console.log("GOLD USD value:", GOLD_THRESHOLD * 2500 / 1e18);

        console.log("\n=== At $3000/ETH ===");
        console.log("BRONZE USD value:", BRONZE_THRESHOLD * 3000 / 1e18);
        console.log("SILVER USD value:", SILVER_THRESHOLD * 3000 / 1e18);
        console.log("GOLD USD value:", GOLD_THRESHOLD * 3000 / 1e18);
    }

    // ============================================================
    // Original Tests (preserved and fixed)
    // ============================================================

    function test_roundZeroExist() public view {
        uint256 currentRound = fundMe.getCurrentRound();
        assertEq(0, currentRound);
    }

    function test_roundZeroNeverUsed() public {
        uint256 currentRound = fundMe.getCurrentRound();
        assertEq(0, currentRound);

        // Open first campaign
        vm.prank(owner);
        fundMe.openCampaign(10e18, block.timestamp + 30 days);

        // Close and withdraw to finalize
        vm.warp(block.timestamp + 31 days);
        // Update price to avoid stale price error
        mockPriceFeed.updateAnswer(int256(ETH_PRICE_MID));
        fundMe.closeCampaign(1);
```

```
        vm.prank(owner);
        fundMe.withdraw(1);

        // Update price again before opening new campaign
        mockPriceFeed.updateAnswer(int256(ETH_PRICE_MID));

        // Now can open new campaign
        vm.prank(owner);
        fundMe.openCampaign(10e18, block.timestamp + 30 days);

        uint256 afterCampaignRound = fundMe.getCurrentRound();
        assertEq(2, afterCampaignRound);
    }
}
```